METRIC

A-A-59596D
30 June 2021
SUPERSEDING
A-A-59596C
4 March 2009

COMMERCIAL ITEM DESCRIPTION

BEARING, BALL, ANNULAR, SINGLE ROW, RADIAL, NON-FILLING SLOT, CARTRIDGE TYPE WITHOUT GREASE PLUG, DIMENSION SERIES 33

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers metric, single row, radial, non-filling slot, cartridge type, annular ball bearings for general-purpose use. These bearings are not intended for use in critical aeronautical or critical special precision applications.

2. CLASSIFICATION. The ball bearings shall be classified by the class, sizes, cage materials, shield and snap ring configurations, precision tolerances, lubricant, and preservative requirements listed below:

Class 9 - dimension series 33

Sizes - bearing dimensions (see Table I)

Cage materials (see Table II)

Shield, seal, and snap ring configurations (see Table III)

Precision tolerances (see Table IV)

Lubricant and preservative requirements (see Table V)

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or DLA Aviation VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil/.

AMSC N/A FSC 3110

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3. SALIENT CHARACTERISTICS

3.1 Dimensions. Bearing boundary dimensions shall conform to the requirements specified in Table I for each of the coded bearing sizes. The listed dimensions conform to the requirements specified for the listed bearing sizes from dimension series 33 in American Bearing Manufacturers Association (ABMA) Standard 20, "Radial Bearings of Ball, Cylindrical Roller, and Spherical Roller Types - Metric Design". The bearing size shall be as specified in the acquisition order (see 7.3(b)). For unlisted bearing sizes, the dimensional and dynamic load rating requirements should also be specified in the acquisition order.

TABLE I. Dimensional requirements (ABMA series 33).

Bearing size code	Bore diameter (mm)	Outside diameter (mm)	Width (mm)	Chamfer r/min. (mm)	Minimum shaft shoulder diameter[1] (mm)	Dynamic load rating (minimum) (lbs.)
04	20	52	22.2	1.1	27	2,760
05	25	62	25.4	1.1	32	3,670
06	30	72	30.2	1.1	37	4,875
07	35	80	34.9	1.5	43.5	5,780
08	40	90	36.5	1.5	48.5	7,075
09	45	100	39.7	1.5	53.5	9,160
10	50	110	44.4	2.0	60	10,730
11	55	120	49.2	2.0	65	12,410
12	60	130	54.0	2.1	72	14,200
13	65	140	58.7	2.1	77	16,000
14	70	150	63.5	2.1	82	18,000
15	75	160	68.3	2.1	87	19,700
16	80	170	68.3	2.1	92	22,700
17	85	180	73.0	3.0	99	22,900
18	90	190	73.0	3.0	104	24,700
20	100	215	82.6	3.0	114	29,900

[1]Listed for reference purposes only. Shoulder height shall be determined to provide sufficient clearance for the direct application of bearing removal force against the bearing inner ring. If the required minimum clearance is not available, an alternative non-destructive bearing removal capability shall be provided.

3.2 Materials.

3.2.1 Rings. The bearing ring material shall be chromium-alloy steel 52100 (UNS G52986) as specified in ASTM International (ASTM) A295/A295M, "Standard Specification for High-Carbon Anti-Friction Bearing Steel". The finished rings shall not exceed the associated billet material inclusion rating also specified in ASTM A295/A295M. Ring hardness shall be at least 58 HRC as defined in ASTM E18, "Standard Test Methods for Rockwell Hardness of Metallic Materials". The grain size of the ring material shall be determined in accordance with ASTM E112, "Standard Test Methods for Determining Average Grain Size".

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3.2.2 Balls. The bearing ball material shall be chromium-alloy steel 52100 (UNS G52986) as specified in ASTM A295/A295M. The finished balls shall not exceed the associated billet material inclusion rating, which is also specified in ASTM A295/A295M. Balls shall be through-hardened no less than 60 HRC, and no more than 67 HRC as defined in ASTM E18. The grain size of the ball material shall be determined in accordance with ASTM E112.

3.2.3 Cage. The bearing cage material shall be impervious to deterioration from any lubricant, preservative, solvent, or other chemical substance expected to contact the bearing during normal use or storage. Similarly, the material shall not cause any chemical deterioration of any other bearing component. Metallic and non-metallic cages shall meet the same bearing performance requirements. Metallic cage materials shall operate from -40 to 250°F (-40 to 121°C). Non-metallic cage materials shall operate from -20 to 250°F (-30 to 121°C). Unless otherwise specified in the acquisition order, the cage material shall be one of the optional coded types listed in Table II (see 7.3(c)).

TABLE II. Cage materials.

Code[1]	Material type
J	Pressed steel
M	Machined bronze or brass
P	Molded plastic (nylon 6/6 or equal)
T	Machined non-metallic (phenolic)
Y	Pressed brass
X	Other (specify in acquisition order)

[1]No significant changes from FF-B-171/10.

3.2.4 Shields and seals. When used, the shield or seal material shall be as recommended by the manufacturer, unless otherwise specified in the acquisition order (see 7.3(d)). The material shall be impervious to deterioration from any lubricant, preservative, solvent, or other chemical substance expected to contact the bearing during normal use or storage. Similarly, the material shall not cause any chemical deterioration of any other bearing component. The shield or seal material shall remain functionally at a minimum temperature range of -20 to 250°F (-30 to 121°C) for non-metallic and -40 to 250°F (-40 to 121°C) for metallic materials. Unless otherwise specified in the acquisition order, the shield configuration for the bearing shall be one of the coded options listed in Table III (see 7.3(e)).

3.2.5 Snap ring. When used, the bearing snap ring and associated snap ring groove shall conform to the requirements specified in ABMA Standard 20.

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TABLE III. Shield, seal and snap ring configurations.

Code[1]	Configuration option
A	Double contact seal
B	Double labyrinth seal
C	Double contact seal with snap ring
D	Double labyrinth seal with snap ring
E	Double shield
F	Double shield with snap ring
X	Other (specify in acquisition order)

[1]No significant changes from FF-B-171/10.

3.3 Precision tolerance. The bearing precision tolerance level shall conform to Annular Bearing Engineers Committee (ABEC) class ABEC-1, ABEC-3, ABEC-5, or ABEC-7 as defined in ABMA Standard 20. The tolerance class and associated radial internal clearance shall be one of the coded options listed in Table IV (see 7.3(f)).

3.3.1 Bearing precision tolerance levels and internal clearances may be affected by the addition of shields and/or seals. Manufacturers/suppliers should be consulted for availability of desired precision tolerances when shields and/or seals are required.

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TABLE IV. Precision tolerance requirements.

Code	Tolerance class	Radial internal clearance
A	ABEC-1	Group 2 (less than normal)
B		Group N (normal)
C		Group 3 (greater than normal)
D		Group 4 (greater than group 3)
T		Group 5 (greater than group 4)
E	ABEC-3	Group 2 (less than normal)
F		Group N (normal)
G		Group 3 (greater than normal)
H		Group 4 (greater than group 3)
U		Group 5 (greater than group 4)
J	ABEC-5	Group 2 (less than normal)
K		Group N (normal)
L		Group 3 (greater than normal)
M		Group 4 (greater than group 3)
V		Group 5 (greater than group 4)
N	ABEC-7	Group 2 (less than normal)
P		Group N (normal)
R		Group 3 (greater than normal)
S		Group 4 (greater than group 3)
W		Group 5 (greater than group 4)

3.4 Lubrication and preservation requirements. Unless otherwise specified in the acquisition order, the bearing lubricant or preservative requirement shall be as selected from the approved coded options listed in Table V (see 7.3(g)). When grease fill is required, the bearing void (the airspace between the inner and outer rings of the assembled bearing) shall be 25 to 50 percent filled. Bearing preservative shall be in accordance with MIL-DTL-197, "Packaging of Bearings, Associated Parts and Subassemblies".

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TABLE V. Lubricant and preservative requirements.

Code	Lubricant or preservative compound
B	Grease in accordance with MIL-PRF-81322
E	Grease in accordance with MIL-PRF-24508
H	Grease in accordance with MIL-PRF-23827
L	Grease with SRI-2 or qualified equivalent
P	Grease in accordance with A-A-59685
S	No fill
T	Preservation compound in accordance with MIL-DTL-197
X	Other (specify in the acquisition order)

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Unless otherwise indicated in the solicitation and resulting contract, the foreign acquisition restrictions in Section 252.225 Clause 252.225.7016, “Restriction on Acquisition of Ball and Roller Bearings,” of the Defense Federal Acquisition Regulation Supplement (DFARS) apply to products described by this CID.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

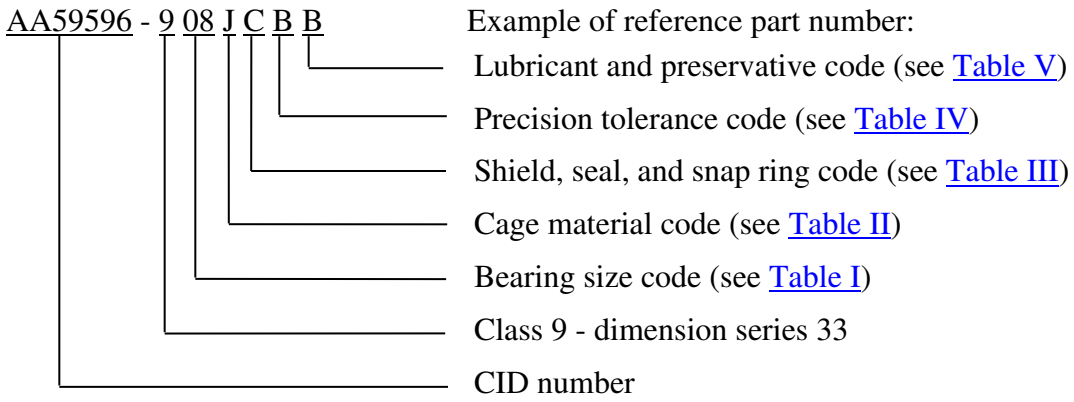
6. PACKAGING

6.1 Preservation, packaging, and marking. Unless otherwise specified in the acquisition order, the bearings supplied shall be preserved, packaged, and marked in accordance with MIL-DTL-197 (see 7.3(h)).

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7. NOTES

7.1 Part or identification number (PIN). The following PIN numbering procedure is for government purposes and does not constitute a requirement for the contractor.



Example of reference part number: AA59596 - 9 08 J C B B identifies a boundary dimension series 33; bore diameter 40 mm, outside diameter 90 mm, width 36.5 mm; pressed steel cage; double contact seal with snap ring; ABEC-1 tolerance class, normal radial internal clearance; filled with grease in accordance with MIL-PRF-81322.

7.2 Sources of documents.

7.2.1 ABMA standards. Copies of ABMA standards may be obtained from the American Bearing Manufacturers Association, 2025 M Street NW, Suite 800, Washington, DC 20036. Electronic copies of ABMA standards may be obtained from https://www.americanbearings.org/.

7.2.2 ASTM standards. Copies of ASTM standards may be obtained from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959. Electronic copies of ASTM standards may be obtained from http://www.astm.org/.

7.2.3 FAR and DFARS. Copies of the FAR and DFARS may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of the FAR and DFARS may be obtained from https://www.acquisition.gov/.

7.2.4 Military specifications and standards. Electronic copies of military specifications and standards may be obtained from https://quicksearch.dla.mil/.

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7.3 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.
b. Bearing size (with dimension/load requirements if size is unlisted) (see 3.1).
c. Cage material type (see 3.2.3).
d. Shield and seal material (see 3.2.4).
e. Shield, seal, and snap ring configuration (see 3.2.4 and 3.2.5).
f. Precision tolerance requirement (see 3.3).
g. Lubricant and preservative requirement (see 3.4).
h. Preservation, packaging, and marking requirements (see 6.1).

7.4 Codes cross-reference. Tables VI, VII, and VIII contain cross-reference data for the part identification number information as listed in FF-B-171/10 and this CID.

7.4.1 Bearing class designations. The CIDs replacing 33 of the 37 associated specification sheets of FF-B-171 have been assigned class codes corresponding to ABMA dimension series. Table VI lists the FF-B-171 specification sheets, the corresponding dimension series, the CID class codes, and the replacement CIDs.

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TABLE VI. Federal specification to CID cross-reference.

FF-B-171 specification sheets	ABMA dimension series	CID class codes	Replacement CID numbers
1	02, 03, 10	1, 2, 4	A-A-59581
2	19	5	A-A-59582
3	10	4	A-A-59583
4	02	1	A-A-59584
5	03	2	A-A-59585
6	04	3	A-A-59586
7	02	1	A-A-59587
8	03	2	A-A-59589
9	32	8	A-A-59595
10	33	9	A-A-59596
11	02	1	A-A-59597
12	03	2	A-A-59598
13	22	6	A-A-59599
14	23	7	A-A-59600
15	19	---	Canceled
16	10	---	Canceled
17	02	---	Canceled
18	03	---	Canceled
19	19	5	A-A-59623
20	10	4	A-A-59624
21	02	1	A-A-59625
22	03	2	A-A-59626
23	04	3	A-A-59627
24	10	4	A-A-59628
25	02	1	A-A-59629
26	03	2	A-A-59630
27	04	3	A-A-59631
28	10	4	A-A-59632
29	02	1	A-A-59633
30	03	2	A-A-59634
31	04	3	A-A-59635
32	32	8	A-A-59636
33	33	9	A-A-59637
34	32	8	A-A-59638
35	33	9	A-A-59639
36	32	8	A-A-59640
37	33	9	A-A-59641

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TABLE VII. Radial internal clearance and ABEC tolerance codes.

FF-B-171/10 codes			A-A-59596 codes		
Code	Radial internal clearance	Tolerance class	Code	Tolerance class	Radial internal clearance
1	Symbol 2	ABEC-1	A	ABEC-1	Group 2
2	Symbol 0		B		Group N
3	Symbol 3		C		Group 3
4	Symbol 4		D		Group 4
			T		Group 5
			E	ABEC-3	Group 2
			F		Group N
			G		Group 3
			H		Group 4
			U		Group 5
5	Symbol 2	ABEC-5	J	ABEC-5	Group 2
6	Symbol 0		K		Group N
7	Symbol 3		L		Group 3
8	Symbol 4		M		Group 4
			V		Group 5
			N	ABEC-7	Group 2
			P		Group N
			R		Group 3
			S		Group 4
			W		Group 5

TABLE VIII. Lubricant and preservative requirements codes.

FF-B-171/10 codes		A-A-59596 codes	
Code	Lubricant or preservative	Code	Lubricant or preservative
A	Grease IAW MIL-PRF-81322	B	Grease IAW MIL-PRF-81322
B	Grease IAW MIL-PRF-24508	E	Grease IAW MIL-PRF-24508
C	Grease IAW MIL-PRF-23827	H	Grease IAW MIL-PRF-23827
D	Grease with SRI-2 or equivalent	L	Grease with SRI-2 or qualified equivalent
	N/A	P	Grease IAW A-A-59685
E	Preservation compound IAW MIL-DTL-197	T	Preservation compound IAW MIL-DTL-197
		S	No fill
		X	Other (specify in the acquisition order)

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7.5 Subject term (key word) listing.

ABEC
cage
precision tolerance
rings

MILITARY INTERESTS:

CIVIL AGENCY
COORDINATING ACTIVITY:
GSA - FAS

Custodians:
Army - AT
Navy - OS
Air Force – 11
DLA - GS

Review Activities:
Army - GL, MI

Preparing Activity:
DLA - GS4

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